

HOW EMS FIRMS CAN SCALE UP To Box Build And Make More Profits



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It is no longer enough to play it safe. EMS firms that step up to full product builds will own the future—one box at a time. The choice is clear: stay stuck, or scale smart.



With the era of electronics manufacturing migrating to India, a new opportunity is fast emerging. However, we see many small and medium electronic manufacturing services (EMS) firms struggling due to pricing issues and shying away from full product builds.

Here are a few ideas that can help you, as the leader of an EMS firm, grow both revenue and margins:

The key shift required is letting go of the low-risk mindset, where customers supply the components and your firm sticks to assembly. This model yields only arbitrage margins, with no manufacturing value add. That said, a hybrid approach may be practical for builds involving high-cost components (such as FPGAs).

Globally, manufacturing costs typically account for 25-30% of the BoM (bill of materials) for standard products. In contrast, regulated sectors—like aerospace, defence, automotive, or medical—can deliver significantly higher margins, often

by at least 10%. Targeting these segments can dramatically increase profitability.

Board assembly alone is heavily machine-driven, and margin expansion is limited, unless your offering includes value-added services, such as PCBA testing, system build, cable harnessing, product testing, certification, calibration, and after-sales support. Defence products, for instance, often generate large annual maintenance contracts, given their lifecycle of 20-25 years.

Most global OEMs expect box-build capabilities. When they begin sourcing in India, they often find a lack of product-level thinking. Since these firms typically cannot train vendors themselves, strong box-building capabilities set you apart from competitors and open doors to multinational clients.

In short, scaling up and improving margins means moving to box build. Here is a list of core skills and competences your team must develop—or you must hire for:

Solid ‘component engineering’ skills. Component engineering is often thought to concern only electronics, but it actually encompasses all product elements—mechanical, plastics, sheet metal, cables, wires, connectors, complex backplanes, etc. Most small or medium EMS players do not have these skills, and scaling to the next level will not be possible unless they develop them.

Post-PCB assembly capabilities.

Trained professionals are needed for post-PCB assembly processes like flying probe testing, in-circuit testing, functional testing, cable harness fabrication, programming devices, and perfectly maintaining product-related manufacturing data, as products in domains like medical, defence, and avionics require meticulous documentation and storage.

Security-focused team and infrastructure. With increasing requirements for embedded cybersecurity in manufacturing, both processes and infrastructure need to be secure, and personnel must be trained and monitored. We know of a company supplying secure devices to the government of India, which has invested in 20% more manpower just to meet these security requirements.

In short, invest in training, develop internal subject-matter experts, and create a solid Learning Management System with a culture of continuous updates. Prioritise video-based training content over static manuals—creating videos is now easier than ever, and they deliver faster, more effective learning outcomes. **EFY**